Open problems:
Schrödinger equation is equivalent to information conservation

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`).

From the brief

1. **Classical counterpart.** State and prove the classical analogue: Hamiltonian evolution is equivalent to preservation of the Shannon entropy and of "the definitional independence of degrees of freedom". (Brief, Sec. 4: "This parallels the classical case where Hamiltonian evolution is equivalent to the preservation of Shannon entropy (Liouville's theorem)".)

2. **Entropy conservation as determinism and reversibility.** The brief reads DR-INFO as "an information theoretic way to characterize determinism and reversibility". A precise definition of deterministic and reversible evolution, and a proof that it is equivalent to DR-INFO, are not given. (Brief, Sec. 1.)

3. **Propagator for unbounded time-dependent $H(t)$.** "The Schrödinger equation gives rise to a differentiable unitary propagator $U_{t_0 \to t_1}$" is assumed. For a continuous family of bounded self-adjoint $H(t)$ this is now proved, in any dimension, from the global existence and uniqueness theorem for linear ODEs: the propagator exists, is unique and is unitary (`exists_propagator_of_continuous`, `propagator_unique`), and in finite dimension its two-parameter evolution satisfies [B]EVMIX, DR-SCEQ, DR-INFO and EVDIFF (`TwoParameter.exists_propagator_isEvolution`; log, "The propagator of a time-dependent Hamiltonian"). Existence, uniqueness and unitarity also hold for bounded $H(t)$ that are only strongly continuous, with $t \mapsto U(t)\psi$ differentiable for every ψ (`exists_propagator_of_continuous_apply`), and so does EVDIFF: $U(t)\rho U(t)^\dagger$ is differentiable in operator norm for every density operator ρ (`hasDerivAt_conj_of_continuous_apply`). The propagator itself need not be norm-differentiable: on $L^2[0,1]$, multiplication by the indicator of $[0,t]$ is such an $H(t)$, and no propagator of it is norm-differentiable on $(0,1)$ (`indicatorHamiltonian_counterexample`). For unbounded self-adjoint $H(t)$ existence needs Kato-type hypotheses and remains open. The converse has the same gap: for unbounded $H(t)$, Step 3 gives only a symmetric $H(t_0)$ (next section, first item). (Brief, Sec. 3, proof of Theorem 2 ($\Rightarrow$); Log, "Differences from the brief".)

From the formalization

Beyond finite dimension the formalization has a part in any dimension; it leaves open the items "Unbounded time-dependent Hamiltonians", "Equation at every time in infinite dimension" and "Wold–Cooper reduction" below. Whether a phase choice makes $t \mapsto U(t)$ differentiable in operator

norm in infinite dimension is settled negatively: on $\ell^2(\mathbb{N})$, a phase e^{it} on the basis vector $e_{\lceil 1/|t| \rceil + 1}$ satisfies all hypotheses, with EVDIFF in operator norm, but no unitary family inducing it is norm-differentiable at 0 (`movingPhase_counterexample`; log, "No norm-differentiable propagator").

1. **Unbounded time-dependent Hamiltonians.** EVDIFF on every density operator forces a bounded H (Banach–Steinhaus). With EVDIFF on a dense set of states, Theorem 2 is formalized for a time-independent evolution, with unbounded self-adjoint H (`drSceq_iff_drInfo_and_evDiff`, by Stone's theorem). Without EVTI, Step 3 gives only a skew-symmetric $A(t_0)$, and $H(t_0)$ need not be self-adjoint: the propagator of $H(t) = H_\beta$ for $t < 0$ and H_α for $t \geq 0$, two self-adjoint extensions of the momentum on $L^2(0,1)$, is differentiable at 0 exactly on the domain of the minimal momentum operator, which is not self-adjoint. Open: the hypotheses (Kato-type, or on the evolution) under which DR-INFO + EVDIFF gives a self-adjoint $H(t)$. The example is formalized: the vectors on which the propagator is differentiable at 0 form a dense subspace, on which $i\hbar U'(0)$ is symmetric but not self-adjoint (`switchedHamiltonian_counterexample`). This subspace is $H_0^1(0,1)$, the domain of the minimal momentum, and $i\hbar U'(0)$ is $-i\,d/dx$ on it (`switchedDomain_eq`). The two pieces of U are $e^{-itH_\alpha/\hbar}$ and $e^{-itH_\beta/\hbar}$ (`twistedTranslation_eq_schrodingerPropagator`). (Log, "Unbounded Hamiltonians".)

   The example uses the two-sided derivative at $t_0 = 0$, and it splits the problem in two. Each one-sided derivative alone has a self-adjoint generator. The right derivative of $e^{-itH_\alpha/\hbar}\psi$ at 0 exists exactly for ψ in the domain of H_α, and the left derivative of $e^{-itH_\beta/\hbar}\psi$ exactly for ψ in the domain of H_β (`exists_hasDerivWithinAt_iff`). The domain of the two-sided generator is the intersection of the two, so it is too small; it is a common restriction of H_α and H_β (`switchedHamiltonian_le_momentum_left`, `switchedHamiltonian_le_momentum_right`).

   (a) *Two-sided derivative*, as in Step 3: DR-INFO + EVDIFF do not give a self-adjoint $H(t_0)$. This is settled by the example above.

   (b) *Forward derivative only.* The evolution $\Phi_{t_0 \to t_1}$ is only given for $t_0 \leq t_1$, so one can define $H(t_0)$ by the right derivative of $t \mapsto \Phi_{t_0 \to t}$ at t_0. The example above says nothing here: its forward generator at 0 is H_α, which is self-adjoint. A continuously varying boundary phase settles it negatively (`sqrtPhase_counterexample`). Take $H(t) = H_{\alpha(t)}$ with $\alpha(t) = \alpha_0 + \sqrt{t}$. Its propagator is unitary and satisfies $U(t_2,t_1)U(t_1,t_0) = U(t_2,t_0)$, and at every $t_0 \neq 0$ the forward generator is the self-adjoint $H_{\alpha(t_0)}$. At $t_0 = 0$ the forward domain is the dense $H_0^1(0,1)$, and the forward generator is the minimal momentum, symmetric but not self-adjoint (`forwardDomain_sqrtPhase`, `not_isSelfAdjoint_forwardHamiltonian_sqrtPhase`). Hölder exponent $\frac{1}{2}$ is the threshold: if $|\alpha(t_0+s) - \alpha(t_0)| = o(\sqrt{s})$, the forward generator is $H_{\alpha(t_0)}$ (`forwardHamiltonian_eq_momentum`). So what is missing is regularity of the evolution in time. Open: a hypothesis on the evolution itself, without reference to a family $H(t)$, under which DR-INFO + EVDIFF give a self-adjoint forward generator. Kato's theorem gives the $\Leftarrow$ direction under its hypotheses (constant domain, $t \mapsto H(t)\psi$ of class C^1); in the example the domain is not constant.

2. **Equation at every time in infinite dimension.** In finite dimension, one differentiable unitary $U(t)$ for all times solves the Schrödinger equation at every time, and $\Phi_{t_0 \to t_1}$ is conjugation by $U(t_1)U(t_0)^\dagger$ (`TwoParameter.drSceq_iff_drInfo_and_evDiff`). The proof fixes the determinant of the local unitaries, which leaves only a root of unity to match when gluing them. In any dimension the determinant is not defined, and the local unitaries are only strongly differen-

tiable, and in general cannot be chosen norm-differentiable (`movingPhase_counterexample`), so the global propagator is not built there. (Log, "The equation at every time".)

3. **Wold–Cooper reduction.** The right shifts V_t on $L^2(0,\infty)$ give an evolution that satisfies [B]EVMIX, maps pure states to pure states, preserves transition probabilities and the entropy, but is induced by no unitary; this is formalized (`shift_semigroup_counterexample`, in the brief "Multiple equivalent characterizations of the Schrödinger equation"). The Wold–Cooper decomposition, which reduces every strongly continuous semigroup of isometries to a unitary group plus copies of this one, is not formalized. (Log, any-dimension part, "Problems found in the brief", second item.)